🚀 Task Board Assessment Challenge

🎯 **Goal:** Build a beautiful, fully-featured Kanban-style task board where users can create tasks, drag them across board sections, and manage work visually — inspired by tools like Asana and Linear.

⏱️ **Time:** 3–6 hours (max 12 hours if you're unfamiliar with the tech stack)

💰 **Cost:** $0 — use free tiers only.

✅ **Required Tech**

- **Supabase** (Free Tier) — database & auth
- **GitHub** (Free Tier) — version control and code sharing

🧩 **Optional Languages / Frameworks**

- React / React Native
- TypeScript
- Go (for backend API, optional)

Use what you're strongest in. You do not need to use all of them.

🎨 **Design — This Is a Priority**

Design quality will be heavily evaluated. We want the app to look and feel polished — not like a generic todo list. Think clean, modern, and intentional.
Take inspiration from tools like **Asana**, **Linear**, and **Notion**. Your board should feel like something a team would actually want to use every day.

Things we're looking for:

- A cohesive color palette and typography system
- Clear visual hierarchy between board sections and task cards
- Smooth drag-and-drop interactions
- Thoughtful empty states, loading states, and error handling
- Mobile-friendly or responsive layout is a plus

🗂️ **The Board (Required)**

Build a **Kanban-style drag-and-drop board** with the following default columns:

- **To Do**
- **In Progress**
- **In Review**
- **Done**

Users must be able to **drag tasks between columns** to update their status. The board should update in real time (or on drop).

🗄️ Database (Supabase)

Create a tasks table with the following fields:

Field	Type	Notes
id	uuid	primary key
title	text	required
status	text	todo, in_progress, in_review, done
user_id	uuid	tied to guest session
created_at	timestamp	auto-set

Bonus fields (recommended):

- description (text)
- priority (low / normal / high)
- due_date (date)
- assignee_id (uuid, references a team member)

Tasks must persist in Supabase with RLS enabled.

👤 Guest Accounts (Required)

Your app must support **guest accounts** so each user only sees their own tasks.

- Use **Supabase Auth anonymous sign-in** (no email/password required)
- Create a guest session automatically on first app launch
- Store each task with a user_id tied to the guest user
- Enable **Row Level Security (RLS)** so users can only read/write their own data

Expected behavior:

- User A creates tasks and only sees User A's tasks
- User B creates tasks and only sees User B's tasks

🖥️ Frontend (Required)

Build a web frontend or React Native app.

Your app must allow users to:

- View their task board with columns
- Create a new task (with a title, and optionally description, priority, due date)
- Drag tasks between columns to update their status
- See loading and error states clearly

⭐ Advanced Features — Highly Recommended if You Want to Stand Out

These features are optional but **strongly encouraged**. They will meaningfully differentiate your submission. Pick any combination:

1. 👥 Team Members & Assignees
- Allow the user to create a small "team" by adding members (name + optional avatar/color)
- Show team members in the board UI
- Let users assign one or more team members to a task
- Display assignee avatars on task cards

2. 💬 Task Comments
- Let users open a task detail panel and write comments
- Show comments in chronological order with a timestamp
- Store comments in a separate comments table in Supabase

3. 📋 Task Activity Log
- Track the history of changes to a task (status changes, edits, assignments)
- Show a timeline of activity inside the task detail view
- Example: *"Moved from To Do → In Progress · 2 hours ago"*

4. 🏷️ Labels / Tags
- Let users create custom labels (e.g. "Bug", "Feature", "Design")
- Assign multiple labels to a task
- Filter the board by label

5. 📅 Due Date Indicators
- Highlight tasks that are due soon or overdue directly on the card
- Show a visual indicator (e.g. colored badge, icon) based on urgency

6. 🔍 Search & Filtering
- Add a search bar to filter tasks by title
- Add filters for priority, assignee, or label

7. 📊 Board Summary / Stats
- Show a small summary: total tasks, tasks completed, tasks overdue
- Could live in a sidebar or header area

🛠️ Backend (Optional)

You may either:
- Call Supabase directly from the frontend, or

- Build a small API (Go optional) with endpoints to get tasks, create tasks, and update tasks

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🌐 Hosting (Required)

Host your frontend so we can use a live demo. Free options:

- Vercel
- Netlify
- Cloudflare Pages

Include the live URL in your final document.

🔐 Access & Sharing

GitHub: Make your repository **public**, or share it as a private repo and send us the link directly. You do not need to add us as a collaborator.

Supabase: Share **read-only access to your database schema** by including the full SQL schema in your final document. You may also share a screenshot of your table structure from the Supabase dashboard. You do not need to add us as a project member.

⚠️ Do not share your Supabase service role key. Use only the public anon key in your frontend. Do not commit secrets to GitHub.

📄 Final Deliverable

Submit one .pdf or .docx file named: firstname_lastname_task_manager_assessment.pdf

Include:

- A short overview of your solution and design decisions
- A link to your **live frontend app**
- A link to your **GitHub repository**
- Your **full database schema** (SQL or clear description)
- Setup instructions to run locally
- Which advanced features you built (if any) and how they work
- Tradeoffs and what you'd improve with more time

✅ What We Evaluate

- Design quality — does it look polished and intentional?
- Board functionality — drag-and-drop, columns, task management
- Frontend usability and state handling

- Database schema and persistence
- Security awareness (RLS, no exposed keys)
- Advanced features (optional but differentiating)
- Code quality and structure
- Live app execution
- Clarity of final document

📤 **Submission:** Upload your PDF or Word document to Google drive and follow the instructions to share the link in the Google Form.